Technical Assessment Report

Problem 1: ESP32 Digital Inputs and Outputs

1. Objective

The aim of this project is to write foundational firmware for an ESP32 microcontroller using the Arduino framework. The task involves configuring digital input pins to read tactile push buttons and digital output pins to control LEDs and a buzzer.

2. Methodology & Logic

To keep the project simple and reliable:

- **Internal Pull-Up Resistors:** The `pinMode(pin, INPUT_PULLUP)` function is used for the buttons. This keeps the pin HIGH normally. When the button is pressed, it connects the pin to GND, making the signal LOW.
- **Outputs:** LEDs and the buzzer are controlled using standard `digitalWrite()` functions (HIGH to turn on, LOW to turn off). Current limiting resistors (220 ohms) are used for the LEDs to prevent burning them out.
- **Debouncing:** Mechanical buttons tend to "bounce" and send multiple signals for a single press. A simple `delay(250)` is used after detecting a button press to ignore these false signals.

3. Pin Configuration

- **Inputs (Active Low):**
    - Button 1 -> GPIO 14
    - Button 2 -> GPIO 27
- **Outputs (Active High):**
    - LED 1 (Green) -> GPIO 25
    - LED 2 (Red) -> GPIO 26
    - Buzzer -> GPIO 33

4. Test Cases & Results

Test Condition	Expected Outcome	Actual Result (Serial Monitor / Hardware)	Status
System Boot	LEDs and Buzzer are OFF. "ESP32 System Ready!" prints on Serial.	LEDs/Buzzer OFF. Message displayed.	PASS
Press Button 1	LED 1 and Buzzer turn ON. Serial prints "Button 1 Pressed".	LED 1 & Buzzer ON. Message displayed.	PASS
Release Button 1	LED 1 and Buzzer turn OFF immediately.	LED 1 & Buzzer OFF.	PASS
Press Button 2	Only LED 2 turns ON. Serial prints "Button 2 Pressed".	LED 2 ON.	PASS
Release Button 2	LED 2 turns OFF immediately.	LED 2 OFF.	PASS

5. Conclusion

The ESP32 was successfully programmed using basic C++ in the Arduino IDE. All inputs were reliably detected using internal pull-up resistors, and outputs behaved exactly as programmed. The simple delay-based debouncing was effective for this level of application.